Redox Reactions

Question1

Consider the following compounds :

$K\underline{O}_2$, $H_2\underline{O}_2$ **and** $H_2\underline{S}O_4$

The oxidation state of the underlined elements in them are, respectively,

NEET 2025

Options:

A. $+1, -2$, and +4

B. $+4, -4$, and +6

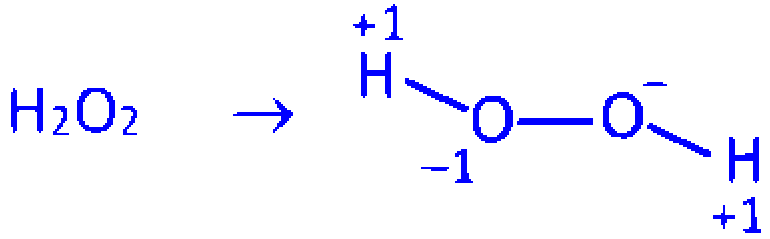
C. $+1, -1$, and +6

D. $+2, -2$, and +6

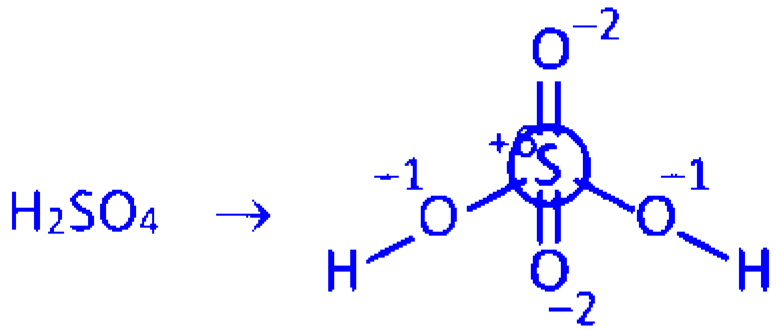
Answer: C

Solution:

$KO_2 \rightarrow$ Alkali metal always shows +1 oxidation state. Therefore oxidation state of K is +1 .



Oxidation state of oxygen in H_2O_2 is -1



Oxidation state of sulphur in H_2SO_4 is +6 .

Question2

The oxidation states not shown by Mn in given reaction is :

$$3MnO_4^{2-} + 4H^+ \rightarrow 2MnO_4^- + MnO_2 + 2H_2O$$

A. +6

B. +2

C. +4

D. +7

E. +3

Choose the most appropriate answer from the options given below :

[NEET 2024 Re]

Options:

A.

D and E only

B.

B and D only

C.

A and B only

D.

B and E only

Answer: D

Solution:

In the following reaction

$$3MnO_4^{2-} + 4H^+ \rightarrow 2MnO_4^- + MnO_2 + 2H_2O$$

Oxidation state of Mn	Species
+6	MnO_4^{2-}
+7	MnO_4^-
+4	MnO_2

So +2 and +3 oxidation state is not shown by Mn.

Question3

Which reaction is NOT a redox reaction?

[NEET 2024]

Options:

A.

$Zn + CuSO_4 \longrightarrow ZnSO_4 + Cu$

B.

$2KClO_3 + I_2 \longrightarrow 2KIO_3 + Cl_2$

C.

$H_2 + Cl_2 \longrightarrow 2HCl$

D.

$BaCl_2 + Na_2SO_4 \longrightarrow BaSO_4 + 2NaCl$

Answer: D

Solution:

Reduction

(1) $\overset{0}{Zn} + \overset{+2}{Cu}SO_4 \longrightarrow \overset{+2}{Zn}SO_4 + \overset{0}{Cu}$, Redox reaction

Oxidation

Reduction

(2) $K\overset{+5}{Cl}O_3 + \overset{0}{I_2} \longrightarrow 2K\overset{+5}{I}O_3 + \overset{0}{Cl_2}$, Redox reaction

Oxidation

Reduction

(3) $\overset{0}{H_2} + \overset{0}{Cl_2} \longrightarrow 2\overset{+1}{H}Cl^{-1}$, Redox reaction

Oxidation

(4) $\overset{+2}{Ba}Cl_2^{-1} + \overset{+1}{Na_2}SO_4^{-2} \longrightarrow \overset{+2}{Ba}SO_4 + 2\overset{+1}{Na}Cl^{-1}$

This is not a redox reaction as there is no change in oxidation state.

Question4

On balancing the given redox reaction,

$$aCr_2O_7^{2-} + bSO_3^{2-}(aq) + cH^+(aq) \rightarrow 2aCr^{3+}(aq) + bSO_4^{2-}(aq) + \frac{c}{2}H_2O(l)$$

the coefficients a, b and c are found to be, respectively-

[NEET 2023]

Options:

A.

3,8,1

B.

1,8,3

C.

8,1,3

D.

1,3,8

Answer: D

Solution:

Solution:

Using Ion electron method:

Reduction Half reaction : $Cr_2O_7^{2-} + 6e^- \rightarrow 2Cr^{3+}$

Oxidation Half reaction: $SO_3^{2-} \rightarrow SO_4^{2-} + 2\bar{e} \times 3$

Overall reaction : $Cr_2O_7^{2-} + 3SO_3^{2-} \rightarrow 2Cr^{3+} + 3SO_4^{2-}$

- To balance 'O' atoms, adding H_2O on LHS

$Cr_2O_7^{2-} + 3SO_3^{2-} \rightarrow 2Cr^{3+} + 3SO_4^{2-} + 4H_2O$

- To balance ' H ' atoms, adding H^+ on RHS

$Cr_2O_7^{2-} + 3SO_3^{2-} + 8H^+ \rightarrow 2Cr^{3+} + 3SO_4^{2-} + 4H_2O$

$\therefore$ $a = 1$

$b = 3$

$c = 8$

Question5

The correct option for a redox couple is :

[NEET 2023 mpr]

Options:

A.

Both are oxidised forms involving same element.

B.

Both are reduced forms involving same element.

C.

Both the reduced and oxidized forms involve same element.

D.

Cathode and anode together.

Answer: C

Solution:

Redox couple is both the reduced and oxidised form involve same element.

Question6

Which of the following reactions is a decomposition redox reaction? [NEET Re-2022]

Options:

A. $P_4(s) + 3OH^-(aq) + 3H_2O(l) \rightarrow PH_3(g) + 3H_2PO_2^-(aq)$

B. $2\,Pb(NO_3)_2(s) \rightarrow 2\,PbO(s) + 4NO_2(g) + O_2(g)$

C. $N_2(g) + O_2(g) \rightarrow 2\,NO(g)$

D. $Cl_2(g) + 2OH^-(aq) \rightarrow ClO^-(aq) + Cl^-(aq) + 4H_2O(I)$

Answer: B

Solution:

Solution:

Lead nitrate decomposed to give PbO, NO_2 and O_2. In this Nitrogen atom oxidation state changes from $+5$ to $+4$ and oxygen changes from -2 to zero.

Question7

What is the change in oxidation number of carbon in the following reaction?
$CH_4(g) + 4Cl_2(g) \longrightarrow CCl_4(l) + 4HCl(g)$
[2020]

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Options:

A. 0 to + 4

B. – 4 to + 4

C. 0 to – 4

D. + 4 to + 4

Answer: B

Solution:

Solution:

(b) $\overset{-4}{C}H_4(g) + 4Cl_2(g) \rightarrow \overset{+4}{C}Cl_4(l) + 4HCl(g)$

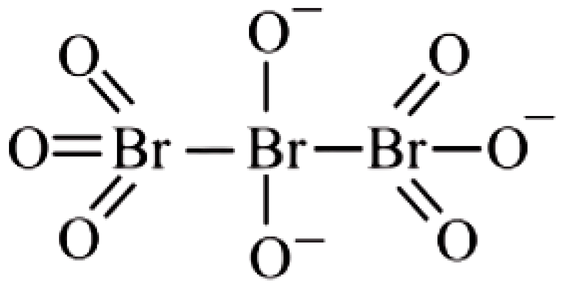
Change in oxidation state of carbon is -4 to $+4$.

Question8

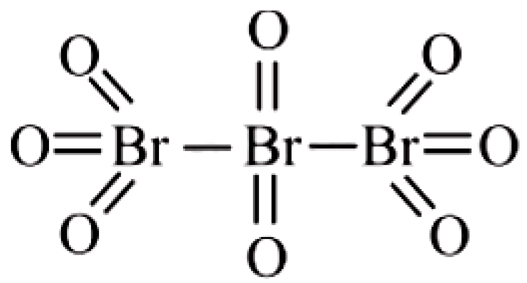
The correct structure of tribromooctaoxide is
(NEET 2019)

Options:

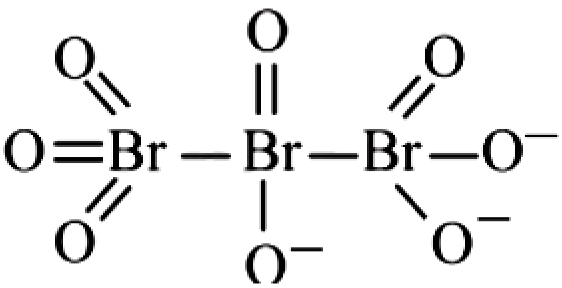
A.



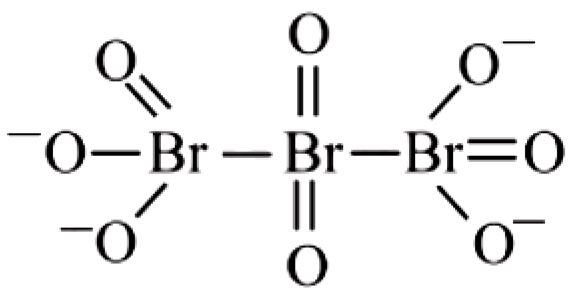
B.



C.



D.



Answer: B

Solution:

Solution:

Question9

Which of the following reactions are disproportionation reactions?
(i) $2Cu^{+} \longrightarrow Cu^{2+} + Cu^{0}$
(ii) $3MnO_4^{2-} + 4H^{+} \longrightarrow 2MnO_4^{-} + MnO_2 + 2H_2O$
(iii) $2KMnO_4 \xrightarrow{\Delta} K_2MnO_4 + MnO_2 + O_2$
(iv) $2MnO_4^{-} + 3Mn^{2+} + 2H_2O \longrightarrow 5MnO_2 + 4H^{+}$
Select the correct option from the following.
(NEET 2019)

Options:

A. (i) and (iv) only

B. (i) and (ii) only

C. (i), (ii) and (iii)

D. (i), (iii) and (iv)

Answer: B

Solution:

Solution:
Disproportionation reactions are those in which the same element\/ compound gets oxidized and reduced simultaneously.
$2Cu^{+} \rightarrow Cu^{2+} + Cu^{0}$
$3MnO_4^{2-} + 4H^{+} \longrightarrow 2\overset{7}{M}nO_4^{-} + \overset{+4}{M}nO_2 + 2H_2O$

Question10

The oxidation state of Cr in CrO_5 is
(Odisha NEET 2019,2014)

Options:

A. -6

B. +12

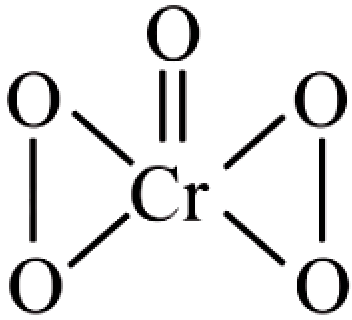
C. +6

D. +4

Answer: C

Solution:

Solution:
CrO_5 has butterfly structure having two peroxo bonds.



Peroxo oxygen has -1 oxidation state. Let oxidation state of Cr be 'x'
$CrO_5 : x + 4(-1) + 1(-2) = 0 \Rightarrow x = +6$

Question11

Consider the change in oxidation state of bromine corresponding to different emf values as shown in the given diagram :

Then the species undergoing disproportionation is (NEET 2018)

Options:

A. BrO_3^-

B. BrO_4^-

C. Br_2

D. $HBrO$

Answer: D

Solution:

Solution:
For a reaction to be spontaneous,
E_{cell}° should be positive.
$HBrO \longrightarrow Br_2 \;\; E^{\circ} = 1.595V$, SRP (cathode)
$HBrO \longrightarrow BrO_3^{-} E^{\circ} = -1.5V$, SOP (anode)
$2HBrO \longrightarrow Br_2 + BrO_3^{-}$
$E_{cell}^{o} =$ SRP (cathode) - SRP (anode)
$= 1.595 - 1.5 = 0.095V$
$E_{cell}^{\circ} > 0 \Rightarrow \Delta G^{\circ} < 0$ (spontaneous)

Question12

The correct order of N -compounds in its decreasing order of oxidation states is
(NEET 2018)

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Options:

A. HNO_3, NO, N_2, NH_4Cl

B. HNO_3, NO, NH_4Cl, N_2

C. HNO_3, NH_4Cl, NO, N_2

D. NH_4Cl, N_2, NO, HNO_3

Answer: A

Solution:

Solution:
$\overset{+5}{HNO_3}, \overset{+2}{NO}, \overset{0}{N_2}, \overset{-3}{NH_4Cl}$

Question13

For the redox reaction,

$$MnO_4^{-} + C_2O_4^{2-} + H^{+} \rightarrow Mn^{2+} + CO_2 + H_2O$$

The correct coefficients of the reactants for the balanced equation are

	MnO^{-}	$C_2O_4^{2-}$	H^{+}
(a)	16	5	2
(b)	2	5	16
(c)	2	16	5
(d)	5	16	2

(NEET 2018)

Options:

A. a

B. c

C. d

D. b

Answer: D

Solution:

Solution:
The correct balanced equation is
$2MnO_4^{-} + 5C_2O_4^{2-} + 16H^{+} \rightarrow 2Mn^{2+} + 10CO_2 + 8H_2O$

Question14

Hot concentrated sulphuric acid is a moderately strong oxidizing agent. Which of the following reactions does not show oxidizing

behaviour?
(NEET- II 2016)

Options:

A.

$Cu + 2H_2SO_4 \rightarrow CuSO_4 + SO_2 + 2H_2O$

B.

$S + 2H_2SO_4 \rightarrow 3SO_2 + 2H_2O$

C.

$C + 2H_2SO_4 \rightarrow CO_2 + 2SO_2 + 2H_2O$

D.

$CaF_2 + H_2SO_4 \rightarrow CaSO_4 + 2HF$

Answer: D

Solution:

Solution:
$CaF_2 + H_2SO_4 \rightarrow CaSO_4 + 2HF$
Here, the oxidation state of every atom remains the same so, it is not a redox reaction.

Question15

(I) $H_2O_2 + O_3 \rightarrow H_2O + 2O_2$
(II) $H_2O_2 + Ag_2O \rightarrow 2Ag + H_2O + O_2$
Role of hydrogen peroxide in the above reactions is respectively (2014)

Options:

A. oxidizing in (I) and reducing in (II)

B. reducing in (I) and oxidizing in (II)

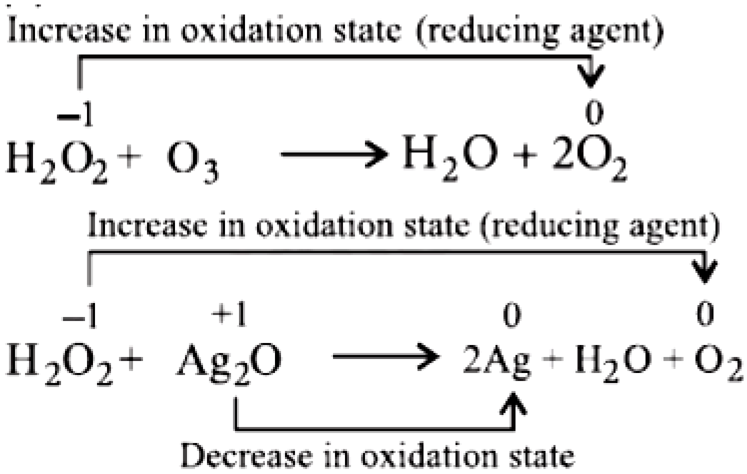
C. reducing in (I) and (II)

D. oxidizing in (I) and (II)

Answer: C

Solution:

Solution:



H_2O_2 acts as reducing agent in all those reaction in which O_2 is evolved

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Question16

The pair of compounds that can exist together is (2014)

Options:

A. $FeCl_3$, $SnCl_2$

B. $HgCl_2$, $SnCl_2$

C. $FeCl_2$, $SnCl_2$

D. $FeCl_3$, KI

Answer: C

Solution:

Solution:
Both $FeCl_2$ and $SnCl_2$ are are reducing agents with low oxidation numbers.

Question17

A mixture of potassium chlorate, oxalic acid and sulphuric acid is heated. During the reaction which element undergoes maximum change in the oxidation number?
(2012)

Options:

A. S

B. H

C. Cl

D. C

Answer: C

Solution:

Solution:
$$\overset{+1}{K}\overset{+5}{Cl}\overset{-2}{O_3} + (COOH)_2 + H_2\overset{+6}{S}O_4 \rightarrow K_2\overset{+6}{S}O_4 + K\overset{-1}{Cl} + CO_2 + H_2O$$
Maximum change in oxidation number of chlorine, i.e., from +5 to -1.

Question18

In which of the following compounds, nitrogen exhibits highest oxidation state?
(2012)

Options:

A. N_2H_4

B. NH_3

C. N_3H

D. NH_2OH

Answer: C

Solution:

Solution:
$N_2H_4 \Rightarrow 2x+4(+1)=0 \Rightarrow 2x+4=0 \Rightarrow x=-2$
$NH_3 \Rightarrow x+3(+1)=0 \Rightarrow x=-3$
$N_3H \Rightarrow 3x+1(+1)=0 \Rightarrow 3x+1=0 \Rightarrow x=-\frac{1}{3}$
$NH_2OH \Rightarrow x+2+1(-2)+1=0 \Rightarrow x+1=0 \Rightarrow x=-1$

Thus,highest oxidation state is $-\frac{1}{3}$

Question19

When Cl_2 gas reacts with hot and concentrated sodium hydroxide solution, the oxidation number of chlorine changes from (2012)

Options:

A. zero to +1 and zero to -5

B. zero to -1 and zero to +5

C. zero to -1 and zero to +3

D. zero to +1 and zero to -3

Answer: B

Solution:

Solution:

$\overset{0}{3Cl_2} + \underset{\text{(hot and conc.)}}{6NaOH} \rightarrow 5Na\overset{-1}{Cl} + Na\overset{+5}{Cl}O_3 + 3H_2O$

This is an example of disproportionation reaction and oxidation state of chlorine changes from 0 to -1 and +5.

Question20

Oxidation numbers of P in PO_4^{3-}, of S in SO_4^{2-} and that of Cr in $Cr_2O_7^{2-}$ are respectively

(2009)

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Options:

A. +3,+6 and +5

B. +5,+3 and +6

C. +3,+6 and +6

D. +5,+6 and +6

Answer: D

Solution:

Solution:

Let oxidation number of P in PO_4^{3-} be x

$\therefore x + 4(-2) = -3 \qquad \Rightarrow x = +5$

Let oxidation number of S in SO_4^{2-} be y

$\therefore y + 4(-2) = -2 \qquad \Rightarrow y = +6$

Let oxidation numbers of Cr in $Cr_2O_7^{2-}$ be z.

$\therefore 2z + 7(-2) = -2 \qquad \Rightarrow z = +6$

Question21

Number of moles of MnO_4^- required to oxidize one mole of ferrous oxalate completely in acidic medium will be

(2008)

Options:

A. 7.5 moles

B. 0.2 moles

C. 0.6 moles

D. 0.4 moles

Answer: D

Solution:

Solution:
$[5e + MnO_4^- + 8H^+ \rightarrow Mn^{2+} + 4H_2O \quad ......(i)] \times 2$
$[C_2O_4^{2-} \rightarrow 2e + 2CO_2 \quad (ii)] \times 5$
$2MnO_4^- + 16H^+ + 5C_2O_4^{2-} \rightarrow 2Mn^{2+} + 10CO_2$
2 moles of MnO_4^- required to oxidise 5 moles of oxalate
∴ no.of moles of MnO_4^- required to oxidise 1 mole of oxalate = 2/5 = 0.4

Question22

Which is the best description of the behaviour of bromine in the reaction given below? $H_2O + Br_2 \rightarrow HOBr + HBr$ (2004)

Options:

A. Proton acceptor only

B. Both oxidised and reduced

C. Oxidised only

D. Reduced only

Answer: B

Solution:

Solution:

$H_2O + \overset{0}{Br_2} \rightarrow H\overset{+1}{OBr} + H\overset{-1}{Br}$

In the above reaction the oxidation number of Br_2 increases from zero (in Br_2) to +1 (in HOBr) and decreases from zero (in Br_2) to -1 (in HBr).

Thus, Br_2 is oxidised as well as reduced and hence it is a redox reaction.

Question23

The oxidation states of sulphur in the anions SO_3^{2-}, $S_2O_4^{2-}$ and $S_2O_6^{2-}$ follow the order

(2003)

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Options:

A. $S_2O_4^{2-} < SO_3^{2-} < S_2O_6^{2-}$

B. $SO_3^{2-} < S_2O_4^{2-} < S_2O_6^{2-}$

C. $S_2O_4^{2-} < S_2O_6^{2-} < SO_3^{2-}$

D. $S_2O_6^{2-} < S_2O_4^{2-} < SO_3^{2-}$

Answer: A

Solution:

Solution:

$SO_3^{2-} : x + (-2)3 = -2$

or $x - 6 = -2$ or $x = +4$

$S_2O_4^{2-} : 2x + (-2)4 = -2$

or $2x - 8 = -2$ or $2x = +6 \quad \therefore x = +3$

$S_2O_6^{2-} : 2x + (-2)6 = -2$

or, $2x - 12 = -2$ or $2x = +10 \therefore x = +5$

Oxidation states follow the order :

$S_2O_4^{2-} < SO_3^{2-} < S_2O_6^{2-}$

Question24

Oxidation state of Fe in Fe_3O_4 is
(1999)

Options:

A. $\frac{5}{4}$

B. $\frac{4}{5}$

C. $\frac{3}{2}$

D. $\frac{8}{3}$

Answer: D

Solution:

Solution:
$Fe_3O_4 \rightarrow 3x + 4(-2) = 0 \Rightarrow x = +\frac{8}{3}$

Question25

Which of the following is redox reaction?
(1997)

Options:

A. Evaporation of H_2O

B. Both oxidation and reduction

C. H_2SO_4 with $NaOH$

D. In atmosphere O_3 from O_2 by lighting

Answer: B

Solution:

Solution:
Redox reactions are those chemical reactions which involve transfer of electrons from one chemical species to another.

Question26

The oxide, which cannot act as a reducing agent is (1995)

Options:

A. CO_2

B. ClO_2

C. NO_2

D. SO_2

Answer: A

Solution:

Solution:
since carbon is in maximum state of $+4$, therefore carbon dioxide (CO_2) cannot act as a reducing agent.

Question27

Which substance is serving as a reducing agent in the following reaction?
$14H^{+} + Cr_2O_7^{2-} + 3Ni \rightarrow 7H_2O + 2Cr^{3+} + 3Ni^{2+}$

(1994)

Options:

A. H^+

B. $Cr_2O_7^{2-}$

C. H_2O

D. Ni

Answer: D

Solution:

Solution:
since the oxidation number of Ni increases from 0 to 2, therefore it acts as a reducing agent.

Question28

The oxidation state of I in $H_4IO_6^-$ is (1994)

Options:

A. +1

B. -1

C. +7

D. +5

Answer: C

Solution:

Solution:

Let $x =$ Oxidation state of I. Since oxidation state of $H = +1$ and oxidation state of $O = -2$, therefore for $H_4IO_6^-$, we get

$(4 \times 1) + x + (6 \times -2) = -1$

or $x = +7$
